

Table 9: A taxonomy of static analysis warnings generated by SonarQube in our dataset.

Category	Description	Example Rules
Code Hygiene	Leftover artifacts from development, including commented-out code, TODO/FIXME comments, unused variables, etc.	S125 (commented code), S1135 (TODO comments), S1128 (unused imports), S1481 (unused variables)
Logic Error	Bugs in code logic, such as incorrect comparisons, unreachable code, infinite loops, and misuse of return values.	S2871 (sort with ill-defined comparator), S1764 (identical sub-expressions), S1763 (unreachable code)
Code Complexity	Structural issues affecting maintainability, including high cognitive complexity, deep nesting, and excessive parameters.	S3776 (cognitive complexity), S2004 (deep nesting), S3358 (nested ternary), S107 (too many parameters)
Accessibility	Web accessibility violations, including missing alt text, keyboard navigation issues, and improper ARIA usage.	S1082 (missing keyboard handler), S5256 (missing table headers), S6848 (non-native interactive elements)
React Patterns	React-specific anti-patterns, including incorrect key usage, hook violations, and state mutation issues.	S6479 (array index as key), S6440 (hook rules violation), S6756 (direct state mutation)
OOP/Design	Object-oriented design issues, including encapsulation violations, improper inheritance, and design pattern misuses.	S1118 (add private constructor), S1104 (public mutable field), S2160 (override equals)
Type Safety	Type system issues primarily in TypeScript, including unnecessary assertions, improper generics, and enum problems.	S4325 (unnecessary type assertion), S6759 (props not readonly), S4621 (duplicate union types)
API Usage	Use of deprecated or outdated APIs and methods that have newer alternatives.	S1874 (deprecated API usage), S6653 (use <code>Object.hasOwn</code>), S6654 (<code>__proto__</code> deprecated)
Error Handling	Incomplete or improper exception and error handling, including empty catch blocks and generic exceptions.	S108 (empty block), S112 (generic exception), S1143 (return in finally), S2737 (empty catch)
Security	Security vulnerabilities, including hardcoded secrets, insecure configurations, and unsafe operations.	S6437 (secrets in image), S2819 (postMessage origin), S5542 (insecure cipher)
Naming Conventions	Violations of language-specific naming conventions for classes, methods, variables, and constants.	S100 (method naming), S101 (class naming), S117 (variable naming)
Regex Issues	Regular expression problems, including excessive complexity, empty matches, and inefficient patterns.	S5843 (regex complexity), S5869 (duplicate char class), S6019 (reluctant quantifier)
Empty/Incomplete	Missing implementations, including empty methods, empty classes, and stub code without logic.	S1186 (empty method), S2094 (empty class), S4658 (empty CSS block)
Concurrency	Threading and asynchronous programming issues, including race conditions and improper synchronization.	S2168 (double-checked locking), S4123 (redundant await), S2696 (non-static in static)
Code Style	Formatting and stylistic preferences, including boolean literals, loop styles, and modern syntax usage.	S1125 (boolean literal), S1301 (switch vs if), S6582 (optional chaining)
Data Science	Python ML/data science specific issues, including random seeds, deprecated NumPy patterns, and DataFrame operations.	S6709 (missing random seed), S6734 (inplace=True), S6730 (deprecated numpy type)
Infrastructure	Docker, Kubernetes, and infrastructure-as-code issues, including image versioning and resource specifications.	S6596 (unversioned image), S6597 (cd vs WORKDIR), S6873 (missing memory request)
Resource Management	Resource lifecycle management issues, including unclosed streams and improper cleanup.	S2093 (try-with-resources), S2095 (unclosed resource), S4042 (<code>Files.delete</code>)
CSS Issues	Stylesheet-specific problems, including invalid properties, unknown units, and duplicate selectors.	S4654 (unknown property), S4653 (unknown unit), S4666 (duplicate selector)
HTML Structure	HTML document structure issues, including missing doctype and improper tag nesting.	DoctypePresenceCheck, MetaRefreshCheck, S4645 (unclosed script)